

Maintenance of the spinning mule at Ulverton

The spinning mule located at the Ulverton historical mill is comprised of gears, rollers, bushings belts, chains, pullies and ropes. Each component must be individually maintained to keep the machine running smoothly. This document serves as a collection of maintenance procedures related to each of the known components and can be expanded if additional components become relevant. An exact maintenance schedule will fall to the employees of the ulverton mill to continuously monitor the mill

Lubricants

Lubrication is the most common form of preventative maintenance, lubricants are used on gears, pins and bushings, chains, pullies, and rollers. There are many different types of lubricants with different applications and purposes. Early lubricants used on the mill were likely to be seed oils, whale oils and later petroleum derivatives. Grease was also used in situations of higher loads and greater temperatures. The link below details a history of lubricants.

<https://mil-comm.com/lubricants/the-ultimate-historical-timeline-of-mechanical-lubrication/>

Gears

Gears are the most common component of the mills main drive head and are the primary way the forces are distributed in the mill. The best practice for the preventative maintenance of gears is a methodical visual inspection before attempting to start the mill. Two links are provided below to aid in the basic maintenance of gears

<https://true-gear.com/basic-gear-maintenance-guidelines/blog.html>

<https://fg-machine.com/blog/do-gears-need-regular-maintenance/>

Pins and bushings

Pins and bushings are the components used to facilitate the connection of a rotational element(gear) to the stationary frame. The pins refer to the rotating rod on which gear, rollers and pulleys are mounted. The bushings refer to the points on the frame that the pins are mounted in, newer machines have replaced bushing with bearings however bushings are still very common. Like gears, pins and bushing should be inspected for wear, pitting and other signs of wear. Bushings are an expected wear item however can be very expensive and difficult to replace. Two links provided below are related to the maintenance of pins and bushings.

<https://www.lubyequipment.com/tips-for-avoiding-wear-on-pins-and-bushings/#:~:text=Rotating%20Pins%20and%20Bushings,around%20the%202%2C000%20hour%20mark.>

<https://www.graco.com/gb/en/vehicle-service/solutions/articles/how-to-avoid-pin-bushing-wear-and-minimize-repair-costs.html#:~:text=The%20importance%20of%20proper%20lubrication,to%20contamination%20%E2%80%93%20resulting%20in%20wear.>

pullies

Pullies and fly wheels follow similar maintenance steps to gears both in terms of inspection and lubrication. Pullies need additional maintenance step depending on the contact method. the mill makes use of belts, ropes, and chains to transfer force. maintaining the main drive belt will depend primarily on the belt material leather the leather should be lightly wiped to remove dust and debris. Maintain the drive system in general is greatly aided by the preventive maintenance manual[1]. Ropes are also used frequently in the mills operation proper maintenance of industrial ropes is less common in modern industry however rope care is verry common in climbing the rope care guide will help keep ropes working longer[2]-[3]. The final pully type used in the ulverton mill is chain and coupling a source has been found to properly maintain these systems and provide relevant context to maintaining all three system types[4]

Belts

[1] https://industrialbeltdrives.com/wp-content/uploads/2017/11/20087_E2_PREVENTIVE_MAINTENANCE_MANUAL.pdf

[2] https://www.marlowropes.com/performance/rope-care-advice/?fbclid=IwZXh0bgNhZW0CMTEAAR2YicMajUGfDvUl3UzMM8_N88mq85hO4XcCqpSQgAPV2m3MjKySWcPFEu4_aem_cZxtfk5DCH_7CkeaNEr8aw

[3]<https://www.machovec.com/rope/care.htm>

[4] <https://reliamag.com/articles/mechanical-maintenance-belt-chain-coupling/>